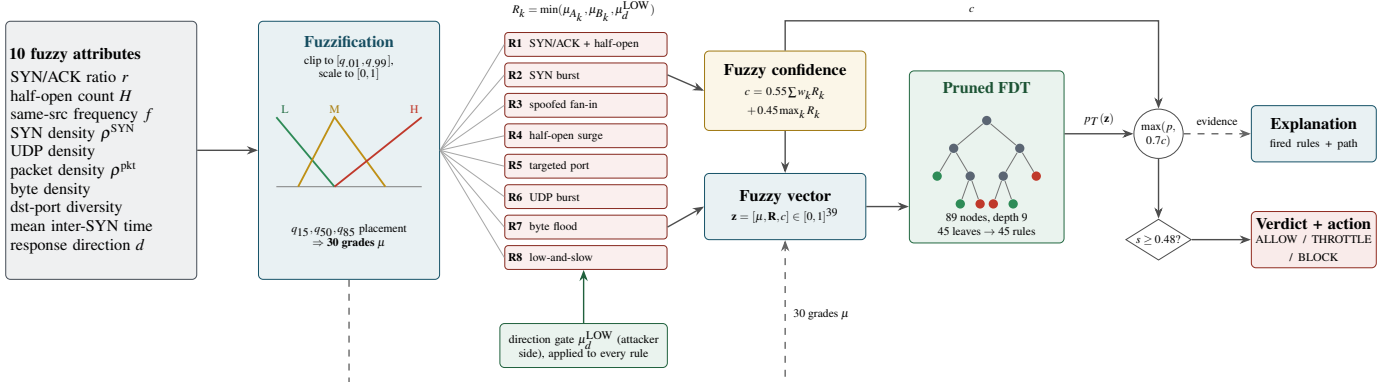

10 fuzzy attributes
SYN/ACK ratio r
half-open count H
same-src frequency f
SYN density ρ^{SYN}
UDP density
packet density ρ^{pkt}
byte density
dst-port diversity
mean inter-SYN time
response direction d
Fuzzification
clip to $[q_{.01}, q_{.99}]$, scale to $[0,1]$
L
M
H
q_{15} q_{50} q_{85} placement
$\Rightarrow$ 30 grades μ
$R_k = \min(\mu_{A_k}, \mu_{B_k}, \mu_d^{\mathrm{LOW}})$
R1 SYN/ACK + half-open
R2 SYN burst
R3 spoofed fan-in
R4 half-open surge
R5 targeted port
R6 UDP burst
R7 byte flood
R8 low-and-slow
direction gate μ_d^{LOW} (attacker side), applied to every rule
Fuzzy confidence
$c = 0.55 \sum w_k R_k + 0.45 \max_k R_k$
Fuzzy vector
$\mathbf{z} = [\mu, \mathbf{R}, c] \in [0,1]^{39}$
30 grades μ
c
Pruned FDT
89 nodes, depth 9
45 leaves $\rightarrow$ 45 rules
$p_T(\mathbf{z})$
$\max(p, 0.7c)$
evidence
Explanation
fired rules + path
$s \geq 0.48$?
Verdict + action
ALLOW / THROTTLE / BLOCK